

Monocular Visual Odometry on Low-Parallax Aerial Imagery: A Comparative Evaluation

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Abstract

We build and evaluate a monocular visual odometry (VO) pipeline on the ALTO aerial dataset, comparing three two-frame pose estimators: SIFT with Essential-matrix decomposition, SIFT with Lucas–Kanade optical flow, and SuperPoint + LightGlue. Hyperparameters are selected by grid search on a held-out validation split, and results are reported on a contiguous test segment. SuperPoint + LightGlue matches the classical estimators on per-pair rotation accuracy, but its trajectory-level drift is roughly an order of magnitude larger, driven by a small number of severe pose failures rather than steady accumulation. We trace these failures to low inter-frame parallax and a dropping Essential-matrix inlier ratio, and show that the domain gap between wide-baseline benchmark datasets (e.g. MegaDepth-1500) and low-parallax, near-nadir aerial imagery reshapes the error distribution, with translation-direction uncertainty becoming the dominant term. We also report the computational footprint of each approach on CPU/GPU.

1 Mathematical Preliminaries

1.1 Two-View Geometry and Pose Recovery

Given two calibrated views with intrinsics K and a set of N corresponding pixel coordinates $\{(p_1^{(i)}, p_2^{(i)})\}_{i=1}^N$, the Essential matrix E satisfies the epipolar constraint

$$\hat{p}_2^{(i)\top} E \hat{p}_1^{(i)} = 0, \quad \hat{p} = K^{-1}p. \quad (1)$$

E factors as $E = [t]_{\times} R$, where $R \in SO(3)$ and $t \in \mathbb{R}^3$ are the relative rotation and translation direction. We estimate E using RANSAC (`cv2.findEssentialMat`, confidence = 0.999) and recover (R, t) via cheirality-checked SVD decomposition (`cv2.recoverPose`).

1.2 Degeneracy Check: Essential vs. Homography

Because a purely planar scene or a purely rotational motion makes E ill-defined, we additionally fit a homography H and compare symmetric transfer scores gated by a χ^2 threshold:

$$S_H = \sum_i [(\tau_H - e_{12}^{(i)})_+ + (\tau_H - e_{21}^{(i)})_+], \quad S_F = \sum_i [(\tau_F - d_{12}^{(i)})_+ + (\tau_F - d_{21}^{(i)})_+], \quad (2)$$

where e_{12}, e_{21} are squared symmetric reprojection errors under H, H^{-1} ($\tau_H = 5.99$, 2 d.o.f., 95%), and d_{12}, d_{21} are squared point-to-epipolar-line distances under F ($\tau_F = 3.84$, 1 d.o.f., 95%). This is the same model-selection score ORB-SLAM uses during monocular initialization to flag near-planar or near-degenerate scenes [5]; we report it here as a diagnostic ratio $R_H = S_H / (S_H + S_F + \varepsilon)$ rather than as a standalone measure of scene planarity.

1.3 Correspondence Retention and Parallax

For each pair we report the fraction of frame-1 keypoints that survive to a valid correspondence after matching or tracking. We call this *correspondence retention* rather than repeatability: in the multi-view geometry literature, repeatability is usually a property of the detector alone (whether a keypoint is independently re-detected under some geometric criterion), whereas the quantity here folds in the matcher or tracker as well, and the exact acceptance criterion differs by estimator — Lowe’s ratio test for SIFT+Essential, optical-flow tracking success for SIFT+LK, LightGlue’s learned match confidence for SuperPoint+LightGlue. The three numbers are therefore comparable as an end-to-end front-end yield, not as a shared geometric property. Median parallax is the median angle, in the camera-1 reference frame, between back-projected inlier bearing vectors:

$$\theta_{\text{par}} = \text{median}_i \arccos(\hat{p}_1^{(i)\top} \hat{p}_2^{(i)}). \quad (3)$$

1.4 Pose Error, RPE and AUC@K

The rotation error (degrees) is

$$e_R = \arccos\left(\frac{\text{tr}(R_{\text{est}} R_{\text{gt}}^\top) - 1}{2}\right) \frac{180}{\pi}. \quad (4)$$

Since monocular translation is scale-free, the translation error is the angle between unit directions:

$$e_t = \arccos(\hat{t}_{\text{est}} \cdot \hat{t}_{\text{gt}}) \frac{180}{\pi}. \quad (5)$$

Following the evaluation protocol established for sparse matching on MegaDepth-1500 [2, 3], we report the Area Under the recall Curve at threshold $K \in \{5^\circ, 10^\circ, 20^\circ\}$ of the maximum angular pose error $e = \max(e_R, e_t)$:

$$\text{AUC@K} = \frac{1}{K} \int_0^K F(\theta) d\theta, \quad F(\theta) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[e^{(i)} \leq \theta]. \quad (6)$$

1.5 Absolute Trajectory Error (ATE)

Relative poses are chained into a trajectory $\{P_k\}$ and aligned to ground truth via the Umeyama closed-form transform [4]. We report two regimes: *direction-only*, where GT scale is injected at every step and the trajectory is aligned rigidly (SE(3), no scale fit, since scale is already correct by construction); and *realistic*, where only the unit-norm translation direction is estimated and a single global scale is fit post-hoc via full Sim(3) alignment. Unqualified “ATE” elsewhere in this note refers to the realistic/Sim(3) regime. To isolate the contribution of rare catastrophic pairs from long-horizon drift, we also compute a **Windowed ATE**, which re-initialises the chain every W steps.

2 Experimental Setup

Dataset. We use a contiguous 200-frame segment of the ALTO dataset [1] (downward-facing RGB video from a helicopter), split chronologically into train/val/test (40/30/30). It is important to note that the train split was utilized *exclusively* during the initial development phase for pipeline debugging and sanity checks. It was not used for hyperparameter tuning, model training, or any form of data-driven optimization. All formal grid searches and final evaluations were conducted strictly on the validation and held-out test splits, respectively.

Camera Model. The camera intrinsic matrix is

$$K = \begin{pmatrix} 324.1 & 0 & 250.0 \\ 0 & 324.1 & 250.0 \\ 0 & 0 & 1.0 \end{pmatrix}.$$

2.1 Estimators and Hyperparameter Tuning

We perform an exhaustive grid search on the validation split to select hyperparameters that minimise the median of $\max(e_R, e_t)$.

Estimator	Grid Searched	Selected
SIFT + Essential	<code>dist_ratio</code> $\in \{0.6, 0.7, 0.8\}$	0.8
	<code>ransac_thresh</code> $\in \{0.5, 0.8, 1.0, 1.5, 2.0\}$	0.5
	<code>nfeatures</code> $\in \{1000, 1500, 2000\}$	2000
SIFT + LK	<code>nfeatures</code> $\in \{1000, 2000, 4000\}$	1000
	<code>ransac_thresh</code> $\in \{0.5, 0.8, 1.0, 1.5, 2.0\}$	0.5
SuperPoint + LightGlue	<code>max_keypoints</code> $\in \{1024, 2048\}$	2048
	<code>ransac_thresh</code> $\in \{0.5, 0.8, 1.0, 1.5, 2.0\}$	0.5

Table 1: Hyperparameter grids and selected values.

Note on RANSAC. The threshold interacts with the feature budget: at larger budgets, higher thresholds consistently scored worse on the validation split. We hypothesize this is because a looser threshold admits more low-quality correspondences into the Essential-matrix estimate on near-degenerate pairs, though we did not run a controlled ablation to confirm this mechanism. The smallest threshold (0.5) was optimal, or tied for best, for all three estimators.

3 Results

3.1 Keypoint Stability and Matching Quality

Estimator	Retent.	Inlier (E)	F Inlier	H Inlier	R_H
SIFT + Essential	0.634	0.916	0.884	0.522	0.595
SIFT + LK	0.968	0.962	0.941	0.627	0.599
SuperPoint + LightGlue	0.730	0.639	0.630	0.284	0.586

Table 2: Correspondence retention and RANSAC-verified matching quality (mean over 56 test pairs).

SIFT + LK retains the largest fraction of correspondences (0.968) and has the highest Essential-matrix inlier ratio. R_H sits in a narrow 0.586–0.599 band across all three estimators, which we read as a property of the scene — the near-nadir, relatively flat ALTO terrain — rather than of the front end.

3.2 Pose Accuracy and Comparison with State-of-the-Art

Estimator	e_R med. ($^\circ$)	e_t med. ($^\circ$)	AUC@5 $^\circ$	AUC@10 $^\circ$	AUC@20 $^\circ$	Acc.@20 $^\circ$
SIFT + Essential	0.177	9.791	0.017	0.107	0.501	96.4%
SIFT + LK	0.139	10.398	0.000	0.085	0.493	98.2%
SuperPoint + LightGlue	0.275	11.857	0.022	0.104	0.377	76.8%
<i>LightGlue</i> [2] (<i>MegaDepth-1500</i>)	—	—	0.561	0.717	0.828	—

Table 3: Relative Pose Error and AUC@ K (test split, gap = 5).

Domain Gap Analysis. The AUC@ K metrics on ALTO fall well short of the MegaDepth-1500 baseline reported in the LightGlue paper [2], and we attribute the gap to scene geometry rather than to matcher quality. MegaDepth is dominated by diverse, high-parallax scene pairs where translation direction is well constrained; ALTO’s near-nadir views have a small baseline relative to scene depth, so translation direction is comparatively hard to pin down (e_t median $\approx 9.8^\circ$ – 11.9° across the three estimators). Since the AUC protocol takes $e = \max(e_R, e_t)$, this alone caps AUC@20 at roughly 50% for both SIFT variants, regardless of their rotational accuracy.

3.3 Geometric Analysis: Drift and Absolute Trajectory Error

Estimator	ATE, dir.-only (m / %)	ATE, realistic (m / %)	s^*
SIFT + Essential	0.042 / 0.35%	0.041 / 0.34%	1.00
SIFT + LK	0.031 / 0.26%	0.031 / 0.26%	1.00
SuperPoint + LightGlue	1.019 / 8.49%	0.730 / 6.08%	1.24

Table 4: ATE over a 12.0 m ground-truth test-split segment (gap = 5).

Estimator	Median Latency (ms)	Median CPU RAM (MB)	Median GPU VRAM (MB)
SIFT + Essential	279.6	27.6	0.0
SIFT + LK	108.6	22.8	0.0
SuperPoint + LightGlue	198.8	0.0	532.6

Table 5: Computational footprint per pair estimation (measured on CPU/GPU; 5 pairs $\times$ 5 runs each, one representative profiling pass — absolute latency varies by a few percent run to run, ranking is stable).

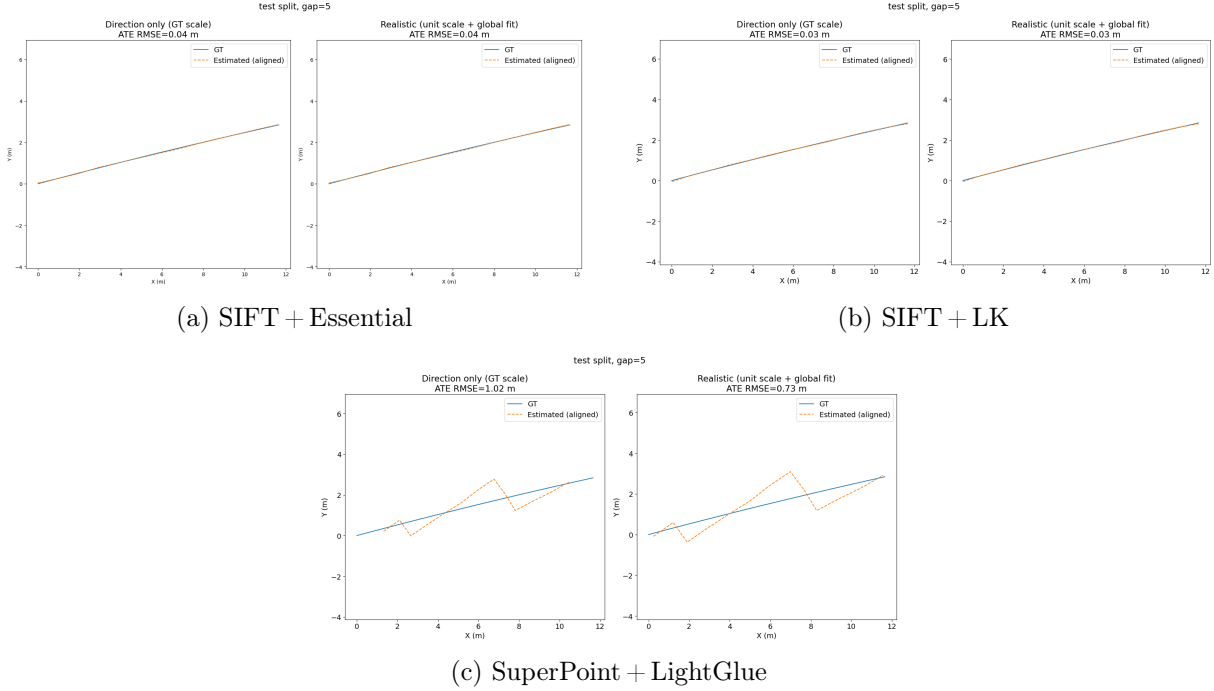


Figure 1: Estimated vs. ground-truth trajectory (direction-only, left panels; realistic scale, right panels). Both SIFT variants overlay almost perfectly on the GT, while SuperPoint + LightGlue exhibits a sharp discontinuity early in the sequence.

SuperPoint + LightGlue matches the classical estimators on per-pair rotation accuracy (Table 3), yet accumulates roughly $18\times$ more absolute trajectory error under the realistic (Sim(3)) alignment, and roughly $24\times$ more under the direction-only alignment (Table 4). The trajectory plots show why: this is not steady drift but a single severe mis-estimation early in the chained sequence that permanently offsets everything downstream of it.

4 Discussion

Estimator Selection and Computational Footprint. Among the methods evaluated on this ALTO test segment, SIFT + LK is the best choice: lowest relative ATE (0.26%), highest correspondence retention (0.968), lowest median latency (108.6 ms), CPU-only, no VRAM. SIFT + Essential trails slightly on accuracy and costs more compute (279.6 ms median). SuperPoint + LightGlue needs a GPU and 532.6 MB of VRAM, and on this dataset that cost buys no accuracy advantage — if anything, it is more exposed than either classical estimator to the low-parallax degeneracies characteristic of near-nadir aerial imagery. We would not generalize this ranking beyond low-parallax, near-nadir aerial footage without re-running the same protocol on a dataset with more viewpoint diversity.

The Translation Direction Bottleneck. The comparison with the LightGlue paper matters mainly as a warning: standard sparse-matching metrics such as $AUC@K$ can be misleading on near-nadir trajectories. The epipolar geometry in this setup produces nearly parallel epipolar lines, so translation direction becomes sensitive to sub-pixel feature-localisation error in a way that has little to do with matcher quality. This caps $AUC@20$ near 50% for both classical methods regardless of their rotational accuracy, so $AUC@K$ alone would rank all three estimators as mediocre even though two of them reconstruct the trajectory almost exactly.

Failure-Rate Analysis and Geometric Degeneracy. To rigorously evaluate robustness, we define a failure as a pair where $e_R > 2^\circ$ or $e_t > 20^\circ$. Under these thresholds, SIFT + LK exhibits a failure rate of only 1.79% (1 out of 56 pairs), and SIFT + Essential 3.57% (2 pairs). In stark contrast, SuperPoint + LightGlue fails in 23.21% of cases (13 out of 56 pairs).

As a diagnostic rather than a causal test, we compute the Spearman rank correlation ρ between the maximum pose error $e = \max(e_R, e_t)$ and each geometric diagnostic, restricted to SuperPoint + LightGlue:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (7)$$

where d_i is the difference in ranks between the pose error and the diagnostic metric. We find moderate-to-strong negative correlations, $\rho(e, \text{parallax}) = -0.64$ and $\rho(e, \text{inlier_ratio}) = -0.48$; with $n = 56$ pairs and only 13 flagged failures, we treat these as suggestive rather than conclusive, and do not report a significance test here (see Limitations). Together with the trajectory plots, they are consistent with SuperPoint + LightGlue’s failures concentrating on low-parallax, low-inlier-ratio frames — a condition that recurs in low-altitude, straight-flight aerial segments — rather than with the scene-planarity statistic R_H , which does not separate the failure cases (Table 2 already shows R_H barely varies across estimators). SIFT + LK is noticeably more resilient on the same pairs.

5 Limitations and Future Work

1. **Unknown Camera-to-Sensor Extrinsic Calibration.** The ground-truth poses are derived from an onboard navigation sensor (e.g., GPS/INS), whereas the visual odometry pipeline estimates the camera pose. The rigid body transformation (extrinsic calibration) between the camera optical center and the navigation sensor frame is not provided in the dataset. This uncalibrated offset introduces a systematic bias, particularly affecting the translation estimates, as the measured translation vector is contaminated by the lever-arm effect during rotational movements. Accounting for this extrinsic offset would likely reduce the residual translation bias further, though we have not established what accuracy floor is actually reachable on this dataset once it is corrected for.
2. **Statistical Power of Rare Events.** The test split yields 56 overlapping pairs. While sufficient to evaluate median drift and windowed ATE, this sample size limits the statistical power of rare-event analysis. For instance, SuperPoint + LightGlue exhibited 13 catastrophic failures under the thresholds $e_R > 2^\circ$ or $e_t > 20^\circ$, corresponding to a failure rate of 23.21%. We treat this as a diagnostic indicator of geometric degeneracy under low parallax rather than a rigorous, population-level statistic. A larger, more diverse aerial dataset is required to confidently quantify the tail of the error distribution.
3. **Absence of Fine-Tuning on the Train Split.** As noted in the experimental setup, the 79-frame train split was reserved solely for code development. All three estimators utilise classical or frozen pretrained pipelines without any adaptation to the target domain. Fine-tuning the LightGlue attention mechanism on the ALTO train split could potentially adapt the matcher to the specific texture and parallax characteristics of aerial nadir views.
4. **RANSAC Variants.** We utilised standard RANSAC. Evaluating LO-RANSAC or MAGSAC++ could further isolate how much of the remaining AUC gap is attributable to the robust estimation backend versus the feature front-end.

References

- [1] I. Cisneros, P. Yin, J. Zhang, H. Choset, S. Scherer. ALTO: A Large-Scale Dataset for UAV Visual Place Recognition and Localization. *arXiv:2207.12317*, 2022.
- [2] P. Lindenberger, P.-E. Sarlin, M. Pollefeys. LightGlue: Local Feature Matching at Light Speed. *ICCV*, 2023.
- [3] Q. Wang. Understanding and Optimizing Attention-Based Sparse Matching for Diverse Local Features. *arXiv:2602.08430*, 2026.
- [4] S. Umeyama. Least-Squares Estimation of Transformation Parameters Between Two Point Patterns. *IEEE TPAMI*, 13(4), 1991.
- [5] R. Mur-Artal, J.M.M. Montiel, J.D. Tardós. ORB-SLAM: A Versatile and Accurate Monocular SLAM System. *IEEE Transactions on Robotics*, 31(5), 2015.